BEING QUASI-AFFINE IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Quasi-compactness and separatedness descend along an fpqc covering. Same chapter, Lemmas lemma-descending-property-quasi-compact and lemma-descending-property-separated, invoked at proof lines 6231-6233. Gives that f is quasi-compact and separated, so that f_* O_X is quasi-coherent.
- The pushforward of the structure sheaf along a quasi-compact separated morphism is a quasi-coherent sheaf of algebras. Schemes, Lemma schemes-lemma-push-forward-quasi-coherent, invoked at proof lines 6236-6237. Makes A = f_* O_X an object whose relative spectrum can be formed.
- The canonical morphism into the relative spectrum, and its base change behaviour. Constructions, Lemma constructions-lemma-canonical-morphism, invoked at proof lines 6238-6242 and 6248-6250. Produces j and identifies its base change with the canonical morphism of f_i.
- Flat base change and the characterisation of quasi-affine morphisms. Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology and Morphisms, Lemma morphisms-lemma-characterize-quasi-affine, invoked at proof lines 6243-6259. Gives g_i^* f_* O_X = f_{i,*} O_{X'} and converts quasi-affineness into the quasi-compact open immersion property of the comparison map.
- Descent of quasi-compact open immersions. Same chapter, Lemma lemma-descending-property-open-immersion, invoked at proof lines 6253-6256. Makes j a quasi-compact open immersion.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01LY: sources/stacks-chow/tags.txt, source lines 2103–2103. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02L3: sources/stacks-descent/pinned-original-source.tex, source lines 6079–6079. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KQ: sources/stacks-descent/pinned-original-source.tex, source lines 5688–5688. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KU: sources/stacks-descent/pinned-original-source.tex, source lines 5798–5798. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01SM: sources/stacks-chow/tags.txt, source lines 2343–2343. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01LC: sources/stacks-chow/tags.txt, source lines 2081–2081. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is quasi-affine" is fpqc local on the base.*

Proof. Let $f : X \to Y$ be a morphism of schemes. Let $\{g_i : Y_i \to Y\}$ be an fpqc covering. Assume that each $f_i : Y_i \times_Y X \to Y_i$ is quasi-affine. This implies that each f_i is quasi-compact and separated. By Lemmas 02KQ and 02KU this implies that f is quasi-compact and separated. Consider the sheaf of $\mathcal{O}_Y$-algebras $\mathcal{A} = f_*\mathcal{O}_X$. By Schemes, Lemma 01LC it is a quasi-coherent $\mathcal{O}_Y$-algebra. Consider the canonical morphism

$$j : X \longrightarrow \underline{\mathrm{Spec}}_Y(\mathcal{A})$$

see Constructions, Lemma 01LY. By flat base change (see for example Cohomology of Schemes, Lemma 02KH) we have $g_i^* f_*\mathcal{O}_X = f_{i,*}\mathcal{O}_{X'}$ where $g_i : Y_i \to Y$ are the given flat maps. Hence the base change j_i of j by g_i is the canonical morphism of Constructions, Lemma 01LY for the morphism f_i. By assumption and Morphisms, Lemma 01SM all of these morphisms j_i are quasi-compact open immersions. Hence, by Lemmas 02KQ and 02L3 we see that j is a quasi-compact open immersion. Hence by Morphisms, Lemma 01SM again we conclude that f is quasi-affine. □